

Joel Daniel Andersson

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Research interests

Differential privacy, online algorithms, streaming algorithms, linear algebra

Current employment

2025-2027 Institute of Science and Technology Austria (ISTA) – Klosterneuburg, Austria
Postdoc in the Monika Henzinger Group
Conducting research on differentially private algorithms.

Education

2022-2025 PH.D. in Computer Science, University of Copenhagen
Thesis title: *On Private Continual Counting* [[link](#)]
Advisor: *Rasmus Pagh*

2014-2020 M.Sc. in Engineering Physics, Lund University
Thesis title: *A Linear Framework for Orbit Correction in the HL-LHC* [[link](#)]
Advisors: *Davide Gamba and Alexandros Sopasakis*

2017-2018 Exchange year. University of California, San Diego
Formally enrolled as a CS student. Studied algorithms, complexity and probability theory.

Professional experience

2020-2022 Ericsson AB – Lund, Sweden
Software Engineer
Implemented protocols in C for the 5G Physical Layer in base stations.

2019-2020 CERN – Geneva, Switzerland
Technical Student (salaried 14-month internship while writing my master's thesis)
Designed a framework in Python for first-order closed orbit analysis in synchrotrons.

2018 CERN – Geneva, Switzerland
openlab Summer Student
Systematized the comparison of beam tracking codes and fixed identified bugs.

2016, 2017 Qlik AB – Lund, Sweden
Summer Software Engineering Intern
Revamped a test framework and automated documentation generation.

Grants, honours & awards

2017 Gull & Stellan Ljungberg Foundation Scholarship
2014 Hvitfeldtska Trust Scholarship
2014 Honorable mention in International Physics Olympiad
2014 Fifth place in Wallenberg (Swedish national) Physics Price Competition

Publications

Authors are listed in alphabetical order by default. First-authors who are first by contribution have a “*” next to their name.

PEER-REVIEWED PUBLICATIONS

- 2026
9. **Improved Accuracy for Private Continual Cardinality Estimation in Fully Dynamic Streams via Matrix Factorization**
J. D. A., Pakak Jain, Satchit Sivakumar.
To appear at Symposium on Principles of Database Systems (PODS), 2026.
[\[arXiv\]](#)
 8. **Learning Rate Scheduling with Matrix Factorization for Private Training**
Nikita Kalinin*, J. D. A.
To appear at Symposium on Foundations of Responsible Computing (FORC), 2026.
[\[arXiv\]](#)
- 2025
7. **Private Lossless Multiple Release**
J. D. A.*, Lukas Retschmeier, Boel Nelson, Rasmus Pagh.
International Conference on Machine Learning (ICML), 2025.
[\[arXiv\]](#) [\[proceedings\]](#) [\[poster\]](#)
 6. **Count on Your Elders: Laplace vs Gaussian Noise**
J. D. A., Rasmus Pagh, Teresa Anna Steiner, Sahel Torkamani.
Symposium on Foundations of Responsible Computing (FORC), 2025.
[\[arXiv\]](#) [\[proceedings\]](#)
 5. **Streaming Private Continual Counting via Binning**
J. D. A., Rasmus Pagh.
IEEE Conference on Secure and Trustworthy Machine Learning (SaTML), 2025.
[\[arXiv\]](#) [\[proceedings\]](#) [\[Google TechTalk\]](#)
- 2024
4. **Continual Counting with Gradual Privacy Expiration**
J. D. A., Monika Henzinger, Rasmus Pagh, Teresa Anna Steiner, Jalaj Upadhyay.
Conference on Neural Information Processing Systems (NeurIPS), 2024.
[\[arXiv\]](#) [\[proceedings\]](#) [\[poster + 5 min video\]](#)

2023 3. **A Smooth Binary Mechanism for Efficient Private Continual Observation**
 J. D. A., Rasmus Pagh.
Conference on Neural Information Processing Systems (NeurIPS), 2023.
[\[arXiv\]](#) [\[proceedings\]](#) [\[poster + 5 min video\]](#)

2019 2. **SixTrack V and Runtime Environment**
 Riccardo De Maria* et al.
International Journal of Modern Physics A 34 (36), 2019.
[\[article\]](#)

1. **SixTrack Version 5: Status and New Developments**
 Riccardo De Maria* et al.
Journal of Physics: Conference Series 1350 (1), 2019.
[\[article\]](#)

NON-PEER-REVIEWED PUBLICATIONS

2025 · **On the Space Complexity of Online Convolution**
 J. D. A., Amir Yehudayoff.
Preprint under review. [\[arxiv\]](#)

2020 · **Orbit Correction Studies on the HL-LHC Layout and Optics V1.5**
 J. D. A., Riccardo De Maria, Davide Gamba.
Technical report, CERN. [\[link\]](#)

Talks

2026 · Presented at the **ISTA TCS Seminar**, March 6, 2026. [\[event link\]](#)
 45 minute talk on *Improved Accuracy for Private Continual Cardinality Estimation in Fully Dynamic Streams via Matrix Factorization*.

2025 · Presenting a paper at **FORC 2025**, June 5, 2025.
 15 minute talk on *Count on Your Elders: Laplace vs Gaussian Noise*.

· Invited talk at the **Boston University Security Seminar**, April 2, 2025.
 45-minute talk on *Streaming Private Continual Counting via Binning*.

· Invited talk at the **Google Privacy Seminar** hosted by Thomas Steinke, February 12, 2025.
 45-minute talk on *Streaming Private Continual Counting via Binning*. [\[video link\]](#)

2023 · Talk at **ARCO 2023**, November 24, 2023.
 20-minute talk on *A Smooth Binary Mechanism for Efficient Private Continual Observation*.

· Invited talk at the **Google Privacy Seminar** hosted by Thomas Steinke, August 30, 2023.
 45-minute talk on *A Smooth Binary Mechanism for Efficient Private Continual Observation*.

Teaching

2022-2025 Teaching assistant for the first-year graduate class in algorithms: Advanced Algorithms and Data Structures (AADS) at the University of Copenhagen. Responsibilities included leading exercise classes, preparing the exam and grading hand-ins. [\[course description\]](#)

Academic service

REVIEWING

ICML'25, PPML@EurIPS'25, ICLR'26, SWAT'26, Journal of Artificial Intelligence Research.

PROGRAM COMMITTEES

SaTML'26 (*Distinguished Reviewer*), TPDP'26.

MISCELLANEA

- 2023 · Organized the biannual ARCO (Algorithmic Research Cooperation around Øresund) event at University of Copenhagen together with Jacob Holm. [[homepage](#)]

Media appearances

- 2024 · Media article(s) on *A Smooth Binary Mechanism for Efficient Private Continual Observation*. English [press release from KU](#), and Danish article in [Dagens Pharma](#).
- 2023 · Contributed to SCIENC KU's *Christmas Calendar*. Recorded a [1-minute video](#), sketching out a connection between Traveling Salesman and Santa's predicament around Christmas.

Other skills

Coding	Python, C, C++, Java, MATLAB, L ^A T _E X, bash, git
Languages	Swedish (<i>native</i>), English (<i>fluent</i>), Danish (<i>B2</i>), French (<i>B1</i>), German (<i>A1</i>)
Running	5 kilometers (19:15), half-marathon (1:35:31), marathon (3:32:57)